# Global Overcomplete Dictionary-Based Sparse and Nonnegative Collaborative Representation for Hyperspectral Target Detection

Chenxing Li, Dehui Zhu, Chen Wu, *Member, IEEE*, Bo Du, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— The combined sparse and collaborative representation (CSCR)-based algorithm is one of the most effective methods among hyperspectral target detection (HTD) methods based on representation and dictionary learning. It encourages target atoms to compete with each other and background atoms to collaborate in the representation. However, this method suffers from several drawbacks. In sparse representation, an overcomplete dictionary is necessary, whereas, in collaborative representation, nonnegative coefficients are required. Besides, the local dual window approach may result in impure background dictionaries obtained from the outer window. To address these issues, we propose a novel approach for HTD, referred to as the global overcomplete dictionary-based sparse and nonnegative collaborative representation (GODSNCR) detector. First, a hierarchical density clustering algorithm is used to complete the dictionary atom extraction to construct a joint overcomplete dictionary to satisfy the dictionary overcompleteness problem required for sparse representation. Second, a nonnegative constraint on the coefficient matrix and a "sum to one" constraint for the joint representation are incorporated to make it more consistent with the physical meaning. Finally, the limitation of the local dual window approach is overcome by substituting the local background dictionary with a global background dictionary. Through the aforementioned strategies, we can use a joint overcomplete dictionary for achieving the sparse representation of targets and utilize a global background dictionary for the collaborative representation of background, the final detection results are obtained by calculating the residuals. The experimental results clearly demonstrate that the proposed algorithm has significant improvement in detection accuracy and strong robustness compared to other typical representation-based HTD methods. Our model will be available at https://github.com/Chenxing-Li/GODSNCR.**

***Index Terms*— Collaborative representation, hyperspectral imagery (HSI), sparse representation, target detection.**

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## I. Introduction

With the development of spectral imaging technology, the spectral resolution of remote sensing images has been dramatically improved. The acquired hyperspectral remote sensing images have the characteristics of high spectral resolution, broadband range, and spectral continuity [1], [2], [3], [4], [5], [6]. Hyperspectral remote sensing images are characterized by a wealth of spectral information [7], [8] covering subtle differences among ground materials. Many substances that are difficult to accurately capture using spatial or broadband information in panchromatic or multispectral remote sensing images can be effectively located and identified in hyperspectral remote sensing [9], [10]. This provides unique advantages for target detection. Due to these advantages, hyperspectral target detection (HTD) has extensive applications in the military, security, environmental monitoring, resource exploration [11], and other fields.

HTD generally refers to separating the desired target from the image background and finding the number and location of target pixels in the hyperspectral image. Specifically, it relies on the Neyman–Pearson principle of the likelihood ratio test, which requires maximizing the probability of target detection while achieving a stable false alarm rate under the constant false alarm rate (CFAR) condition [12], [13].

Since the 1990s, traditional methods such as subspace projection method [14], [15], spectral matching method [16], [17], and probabilistic statistical model [18], [19], [20], [21] have been employed for HTD. Due to the complexity of hyperspectral images, the aforementioned classic methods for HTD often rely on specific statistical assumptions. Consequently, they only achieve satisfactory detection results under certain conditions.

Subsequently, with the development of machine learning and deep learning, various methods have emerged in the territory of HTD. In [22], [23], and [24], the kernel spectral matched filter (KSMF), the kernel constrained energy minimization (KCEM), the kernel orthogonal subspace projection

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(KOSP) are proposed as target detectors based on kernel technique. Methods based on representation and dictionary learning, such as the sparsity-based target detector (STD) [25], the combined sparse and collaborative representation (CSCR) detector [26], and the sparse representation-based binary hypothesis (SRBBH) [27] detector, use a linear combination of elements in the dictionary to reconstruct the pixels in the hyperspectral image and then utilize these reconstructed pixels for detection. Other newly proposed methods based on representation and dictionary learning continue to follow this concept. For example, the collaborative-guided spectral abundance learning (CGSAL) model [28] detects subpixel targets by learning a spectral abundance of the target signature in each pixel. These representation and dictionary-based learning methods have overcome the limitations of traditional approaches that rely on complex mathematical models and manual feature selection. By considering the spectral sparsity of hyperspectral images, these methods enhance the expressive power and adaptability of features while reducing feature redundancy and dimensionality. As a result, they enable more precise target identification and separation from the background in hyperspectral remote sensing imagery, exhibiting superior detection performance. Consequently, representation and dictionary-based learning methods have become one of the mainstream approaches in the field.

Given the advent of deep learning, the utilization of deep feature extraction in HTD has also gained prominence. In recent years, deep neural networks have led to remarkable success in high-level feature extraction for the effective characterization of nonlinear hyperspectral images. One popular approach is to construct deep learning networks for HTD using a binary classification concept. Examples include [29], [30], [31], and [32], which all employ this strategy. They build positive sample pairs between target samples and negative sample pairs between target samples and background samples for network training, thereby accomplishing the binary classification task. Specifically, for example, in [29], a method denoted as HTD-net adopts the idea of U-net and proposes a U-autoencoder to generate training samples and construct sample pairs to accomplish target detection. Another approach is to detect by reconstructing hyperspectral images, such as [33], [34], [35], and [36] reconstruct the original hyperspectral images to eliminate redundant information and enhance useful information. Reconstruction methods often employ generative models like autoencoders or generative adversarial networks. Ultimately, the final detection results are obtained by comparing the reconstructed images with the original hyperspectral images. For example, the background learning model known as the target suppression constraint-based background learning (BLTSC) [33] model achieves the reconstruction of the background through the adversarial autoencoder, and completes target detection by comparing the difference between the reconstructed hyperspectral image and the original image. In addition to the two main methods mentioned above, there are also some other novel approaches that construct networks for target detection using different strategies. The HTD based on interpretable representation network (HTD-IRN) [37] constructs a subspace representation network to separate the background and the target components, within which the background subspace can learn adaptively. By minimizing the background output capability and maximizing the target response to construct the loss function, the detection of targets is fulfilled effectively.

The research in this article will continue to advance in the direction of representation and dictionary-based methods. Generally, representation and dictionary-based learning methods complete the target detection by representing the test pixels using a background dictionary and a target dictionary. Then, the detection result is obtained by calculating the residuals between the two representation processes [38]. Within such methods, the CSCR detector [26] is often used as one of the benchmark methods. The CSCR detector takes into account that individual pixels in hyperspectral imagery (HSI) exist in a low-dimensional subspace. Consequently, it represents the target by sparsely combining training samples in a linear manner, promoting competition among dictionary atoms. In addition, it considers the possibility of multiple background atoms within a single pixel and employs a collaborative representation of the background, prioritizing the utilization of all available atoms. Finally, the detection result is obtained by utilizing the residuals derived from the two representations, thereby achieving an accurate and effective target detection.

However, the CSCR detector still encounters several challenges. First, the sparse representation part uses only a few prior spectra to construct the target dictionary, which fails to meet the criteria of an overcomplete dictionary necessary for achieving sparse representation [39]. Second, the collaborative representation coefficients often have negative terms in the actual solution process, which do not meet the actual physical meaning. Finally, the local dual window approach may introduce impurities into the background dictionary due to the presence of targets, noise, or abnormal disturbances in the outer window [40].

In an effort to alleviate these challenges, this article proposes a global overcomplete dictionary-based sparse and nonnegative collaborative representation (GODSNCR) detector. This article offers the following notable contributions.

1) We employ a hierarchical density clustering algorithm to extract dictionary atoms and construct an overcomplete dictionary in the dictionary construction phase that fulfills the requirement of dictionary overcompleteness for sparse representation.
2) A nonnegative constraint and a "sum to one" constraint are imposed on the coefficient matrix to address the problem of negative coefficients in the collaborative representation that does not satisfy the actual physical meaning of the collaborative representation.
3) We use the global background dictionary obtained by hierarchical density clustering instead of the local background dictionary constructed by the local dual window approach to solve the problem of impurity of the background dictionary due to the presence of targets, noise, or anomalous interference in the outer window.

The remainder of the article is structured as follows. Section II offers a comprehensive explanation of the proposed

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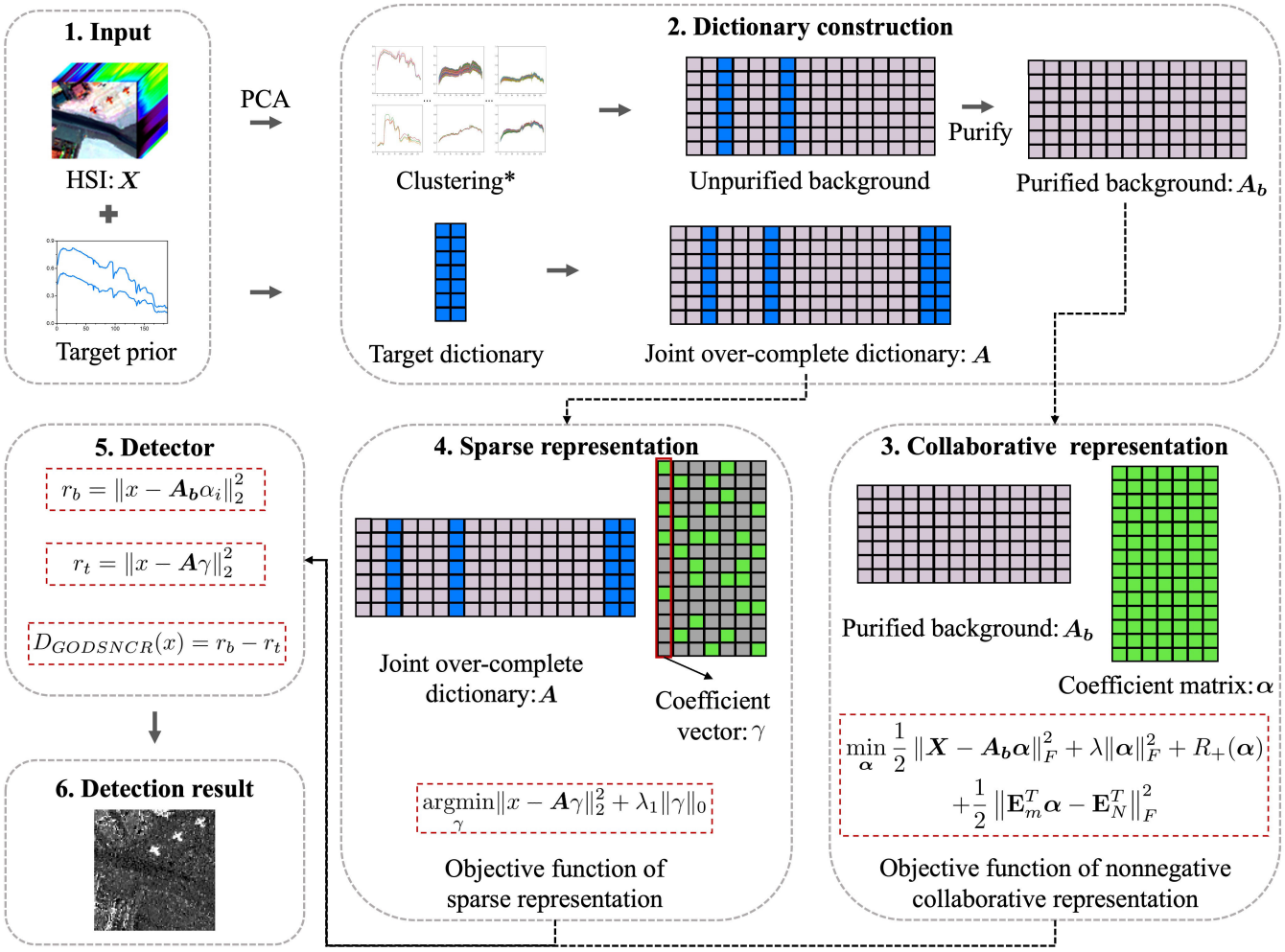



Fig. 1. Flowchart the GODSNCR algorithm. *The clustering results are presented according to the data form before dimensionality reduction. All subsequent data processing uses the data before dimensionality reduction.

GODSNCR detector. Section III demonstrates experimental results. Finally, concluding remarks are provided in Section IV.

## II. Proposed Methodology

This section provides a comprehensive description of the GODSNCR detector. The overall algorithm flow can be illustrated by Fig. 1. As depicted in Fig. 1, the entire process can be divided into six parts, including data input, dictionary construction, collaborative representation, sparse representation, detector design, and finally obtaining the detection results. Specifically, we first reduce the dimension of the input hyperspectral data. Then, we employ a clustering algorithm to extract the categories of ground objects and extract dictionary atoms from the clustering results, constructing an unpurified background dictionary. By combining the dictionary with prior targets, the joint overcomplete dictionary can be constructed for sparse representation, and the purified background dictionary can be obtained for collaborative representation. After the sparse and collaborative representation coefficients are obtained, respectively, the detection values are obtained through the residual calculation under the two representations. The detailed design of each part of the algorithm is presented in Sections II-A–II-C.

### A. Dictionary Construction

After inputting the hyperspectral data, we initiate the construction of a global overcomplete dictionary. In the dictionary construction process, we need to select a certain number of dictionary atoms from the hyperspectral data without any background prior knowledge, and these dictionary atoms can cover all the features in the image.

Therefore, we utilize clustering, an unsupervised learning algorithm, to process the data, complete the extraction of data categories, and select dictionary atoms proportionally from each cluster after data grouping to complete the construction of a global overcomplete dictionary.

*1) Clustering:* Typically, for high-dimensional data like hyperspectral data, dimension reduction is commonly performed. Through this process, it facilitates the reduction of noise and redundant information and improves the validity of distance or similarity. Principal component analysis (PCA) [41] algorithm is a commonly used method for dimension reduction of data, in order to improve the accuracy of clustering results, PCA is employed to perform dimensionality

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reduction on hyperspectral data. It should be noted that PCA is solely applied in the construction of the background dictionary. As the construction of the background dictionary necessitates pure background samples uncontaminated by target interference, better results are achieved by excluding targets from the construction of the background samples. However, in the processes of sparse representation, collaborative expression, and subsequent detector construction, the original signals are utilized without undergoing PCA dimensionality reduction. Therefore, targets are not removed during these stages. Based on the experience of previous studies [42], [43], [44], [45], [46], in order to reduce computational costs while ensuring the preservation of principal component information, we choose to reduce the dimensionality to 30. After dimension reduction is completed, it is crucial to select the proper clustering algorithm to complete the next step. Here, we need to consider some relevant factors and requirements in clustering: a) the spectra of the background and the number of background categories are unknown, which mean we need to complete the clustering when the number of clustering clusters is unknown; b) the atoms in the background dictionary ought to cover all the constituent elements of the ground object in the background as comprehensively as possible, which mean that we need to make the cluster of clustering results sensitive to the less distributed categories; and c) reduce the interference of noise on the dictionary, which mean we need to choose a clustering algorithm with low sensitivity to noise to make the clustering result pure. In summary, we choose a density-based clustering algorithm to extract dictionary atoms. Among these algorithms, the density-based spatial clustering of applications with noise (DBSCAN) [47] algorithm is a widely utilized choice. However, it necessitates setting the neighborhood radius (Eps) and the minimum number of samples (MinPts) in the neighborhood, both of which largely influence the clustering results and can be hard to determine. Furthermore, the DBSCAN algorithm assumes uniform densities among all clusters, which may not hold true in practical scenarios.

The improved form of the DBSCAN algorithm: hierarchical DBSCAN (HDBSCAN) [48] is easier to select parameters than the DBSCAN algorithm. It is only necessary to select the minimum cluster size (min _cluster_size) and the number of samples near a point that are considered to be core points (min _samples) within the integer range. Moreover, HDBSCAN can deal with clusters of different densities. By constructing minimum spanning trees and hierarchical clustering structures, HDBSCAN can identify and extract clusters at different density levels more easily. Therefore, we choose to use the HDBSCAN algorithm to complete the clustering work.

As for the distance measure for clustering, the Euclidean distance, which is easy to compute, is directly adopted here.

*2) Selection and Purification of Clustering Results:* After completing the extraction of clusters in the hyperspectral dataset, we select the most representative points from each cluster, namely, the feature points of the cluster, according to the proportion of background dictionary atoms. They are selected according to the stability of points in the HDBSCAN algorithm (refers to the degree to which a point stays in the same cluster during the splitting process, and depends on the density of points and clusters). If the feature points obtained by the proportional selection of a cluster are insufficient (which is rare), other points of the cluster are randomly selected after all feature points of the cluster are selected.

After the background atoms are selected in this manner, an unpurified overcomplete dictionary is constructed, which contains a few number of target samples from the target cluster obtained by clustering. The unpurified overcomplete dictionary combined with the prior target can form a joint overcomplete dictionary $\boldsymbol{A}$ for the sparse representation of the target. In order to construct a pure background dictionary, it is necessary to purify the unpurified overcomplete dictionary to filter out target atoms in the dictionary. Specifically, by comparing the prior target spectra, the closest atoms that are a certain proportion away from the prior target spectra (5% according to experience) are filtered out. The purified background dictionary $\boldsymbol{A_b}$ is obtained, which can be used for the collaborative representation of background. Thus, the construction of a global overcomplete dictionary is completed.

### B. Background Consistency Constraints

Once the dictionaries are constructed, it is ready for sparse and collaborative representation. During the process of collaborative representation, coefficients for some collaborative representation terms often exhibit negative values. The negative coefficients are not consistent with the actual physical meaning and also have a negative influence on the results. Therefore, the solution procedure needs to be constrained. Here, the nonnegative constraint and the "sum to one" constraint are discussed. The constraint-solving procedure is detailed below.

*1) Collaborative Representation Constraint:* For the collaborative representation of the background, we need to use the purified background dictionary obtained in the previous step for the collaborative representation of all the pixels to be tested. Different from other local algorithms, we conduct a collaborative representation solution for all pixels simultaneously for convenience and operational efficiency improvement since a global background dictionary is used here.

Assuming there are $m$ background atoms and $N$ test pixels in the task, and each pixel has L-bands. The objective function for the collaborative representation of the background can be written as follows:

$$\min_{\boldsymbol{\alpha}} \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A_b}\boldsymbol{\alpha}\|_F^2+\lambda\|\boldsymbol{\alpha}\|_F^2+R_+(\boldsymbol{\alpha})+\frac{1}{2}\left\|\mathbf{E}_m^T\boldsymbol{\alpha}-\mathbf{E}_N^T\right\|_F^2 \quad (1)$$

where $\boldsymbol{X}$ denotes the matrix of image elements to be measured, $\boldsymbol{X} \in \mathbb{R}^{L\times N}$, $\boldsymbol{A_b}$ denotes the background dictionary, $\boldsymbol{A_b} \in \mathbb{R}^{L\times m}$, $\boldsymbol{\alpha} \in \mathbb{R}^{m\times N}$ is the coefficient matrix, $1/2\|\mathbf{E}_m^T\boldsymbol{\alpha}-\mathbf{E}_N^T\|_F^2$ is the "sum to one" constraint on the coefficient matrix, where $\mathbf{E}_m$ and $\mathbf{E}_N$ are the column vectors with all elements being 1, $\mathbf{E}_m \in \mathbb{R}^{m\times 1}$, $\mathbf{E}_N \in \mathbb{R}^{N\times 1}$, $R_+(\cdot)$ is a nonnegative constraint, where

$$R_+(\boldsymbol{\alpha}(i,j)) = \begin{cases} 0, & \boldsymbol{\alpha}(i,j) \geq 0 \\ +\infty, & \boldsymbol{\alpha}(i,j) < 0. \end{cases} \quad (2)$$

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*2) Optimizing Solutions:* In the previous summary, the objective function is proposed as a convex function with an inequality constraint. In this article, the alternating direction method of multipliers (ADMM) [49] is employed to achieve an optimal solution to the objective function. First, the slack variable $\boldsymbol{Z}$ is introduced into (1), which is then reformulated as follows:

$$\min_{\boldsymbol{\alpha},\boldsymbol{Z}} \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A_b}\boldsymbol{\alpha}\|_F^2+\lambda\|\boldsymbol{\alpha}\|_F^2+R_+(\boldsymbol{Z}) + \frac{1}{2}\left\|\mathbf{E}_m^T\boldsymbol{\alpha}-\mathbf{E}_N^T\right\|_F^2 \text{ s.t. } \boldsymbol{Z}=\boldsymbol{\alpha}. \quad (3)$$

Next, the Lagrange multiplier $\boldsymbol{W}$ is introduced, and (3) is rewritten as the extended Lagrange equation [50]

$$\mathcal{L}_\rho(\boldsymbol{\alpha},\boldsymbol{Z},\boldsymbol{W}) = \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A_b}\boldsymbol{\alpha}\|_F^2+\lambda\|\boldsymbol{\alpha}\|_F^2+R_+(\boldsymbol{Z}) + \frac{1}{2}\left\|\mathbf{E}_m^T\boldsymbol{\alpha}-\mathbf{E}_N^T\right\|_F^2+\langle\boldsymbol{W},\boldsymbol{\alpha}-\boldsymbol{Z}\rangle + \frac{\rho}{2}\|\boldsymbol{\alpha}-\boldsymbol{Z}\|_F^2 \quad (4)$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product of the matrix and $\rho$ represents the penalty term parameter that affects the convergence speed of the ADMM algorithm (a large number of experiments indicate that setting $\rho$ to 0.05 results in better performance).

Subsequently, the optimization problem can be decomposed into several subproblems, and by iteratively solving these subproblems sequentially, the solution to the original optimization problem can be achieved. The obtained subproblems are

$$\boldsymbol{\alpha} \leftarrow \min_{\boldsymbol{\alpha}} \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A_b}\boldsymbol{\alpha}\|_F^2+\lambda\|\boldsymbol{\alpha}\|_F^2 + \frac{1}{2}\left\|\mathbf{E}_m^T\boldsymbol{\alpha}-\mathbf{E}_N^T\right\|_F^2+\frac{\rho}{2}\left\|\boldsymbol{\alpha}-\boldsymbol{Z}+\frac{\boldsymbol{W}}{\rho}\right\|_F^2 \quad (5)$$

$$\boldsymbol{Z} \leftarrow \min_{\boldsymbol{Z}} R_+(\boldsymbol{Z})+\langle\boldsymbol{W},\boldsymbol{\alpha}-\boldsymbol{Z}\rangle+\frac{\rho}{2}\|\boldsymbol{\alpha}-\boldsymbol{Z}\|_F^2 \quad (6)$$

$$\boldsymbol{W} \leftarrow \min_{\boldsymbol{W}}\langle\boldsymbol{W},\boldsymbol{\alpha}-\boldsymbol{Z}\rangle. \quad (7)$$

The iterative update process of $\boldsymbol{\alpha}$ is obtained by taking the partial derivative of $\boldsymbol{\alpha}$ through (5) and making its partial derivative zero as follows:

$$\boldsymbol{\alpha}_{t+1} = \left(2\lambda\mathbf{I}+\boldsymbol{A}_{\boldsymbol{b}}^T\boldsymbol{A_b}+\mathbf{E}_m\mathbf{E}_m^T+\rho\mathbf{I}\right)^{-1} \times\left(\boldsymbol{A}_{\boldsymbol{b}}^T\boldsymbol{X}+\mathbf{E}_m\mathbf{E}_N^T+\rho\boldsymbol{Z}-\boldsymbol{W}\right). \quad (8)$$

Optimizing the solution for the nonnegative variable $\boldsymbol{Z}$ in (6) by forcing the solution into the nonnegative quadrant yields [51]

$$\boldsymbol{Z}_{t+1} = \max(\boldsymbol{\alpha}_{t+1}+\boldsymbol{W}_t/\rho, 0). \quad (9)$$

Optimization of the Lagrange multiplier in (7) can be solved by gradient ascent to obtain

$$\boldsymbol{W}_{t+1} = \boldsymbol{W}_t+\rho(\boldsymbol{\alpha}_{t+1}-\boldsymbol{Z}_{t+1}). \quad (10)$$

The termination condition of the above iterative process is that the iterations reach the maximum number of iterations $t_{\max}$ or the relative tolerance $\epsilon \le 10^{-2}$, and the termination condition is expressed as

$$\varepsilon = \left\|\boldsymbol{\alpha}^{t+1}-\boldsymbol{\alpha}^t\right\|_F/\left\|\boldsymbol{\alpha}^t\right\|_F \le 10^{-2}. \quad (11)$$

### C. Design of the Proposed GODSNCR

After obtaining the coefficient matrix $\boldsymbol{\alpha}$ of the background collaborative by the method explained in Section II-B, each column $\alpha_i$ of the coefficient matrix is the collaborative coefficient of the corresponding image element $x$ to be measured. Therefore, the corresponding background reconstruction residual can be calculated as

$$r_b = \|x-\boldsymbol{A_b}\alpha_i\|_2^2. \quad (12)$$

After solving the collaborative background representation through background constraints, the joint overcomplete dictionary $\boldsymbol{A}$ obtained in section A is used for the sparse representation of the target. A test pixel $x$ can be represented by the joint overcomplete dictionary $\boldsymbol{A}$ whose columns are the background and target training samples. By utilizing the training sample dictionary, we can obtain a representation that satisfies the equation $\boldsymbol{A}\gamma = x$ by addressing the optimization problem stated below to find the sparsest vector [25], [30], [42]

$$\underset{\gamma}{\text{argmin}}\|x-\boldsymbol{A}\gamma\|_2^2+\lambda_1\|\gamma\|_0. \quad (13)$$

By the orthogonal matching pursuit (OMP) [52] algorithm, the sparse representation coefficient $\gamma$ of each pixel is obtained, and the reconstruction residual of the target can be obtained by sparse reconstruction as

$$r_t = \|x-\boldsymbol{A}\gamma\|_2^2. \quad (14)$$

Therefore, the target detector of the final proposed method is

$$D_{\text{GODSNCR}}(x) = r_b-r_t. \quad (15)$$

The algorithmic details of the whole target detection process are shown in Algorithm 1.

## III. Simulation Results and Analysis

For the purpose of demonstrating the effectiveness of the GODSNCR algorithm, we meticulously perform extensive experiments on three real HSIs that are publicly available. The experiments are conducted on a PC with the following specifications: Intel[1] Xeon[1] CPU E3-1240 V5.5 at 3.50 GHz, 16 GB, and 64-bit Windows 10 operating system. Especially, two deep learning-based methods, HTD and HTD-IRN, are carried out in Python 3.9 on GTX 3090 GPU. Subsequently, we will provide comprehensive details on the experimental data, the experimental setup, the results, and the analysis of target detection, as well as the parameter sensitivity analysis.

### A. Hyperspectral Datasets

To assess the target detection performance of the GODSNCR algorithm, this article employs three publicly available datasets: Mosaic, RIT Campus, and AVIRIS datasets. Detailed information regarding these datasets is presented below.

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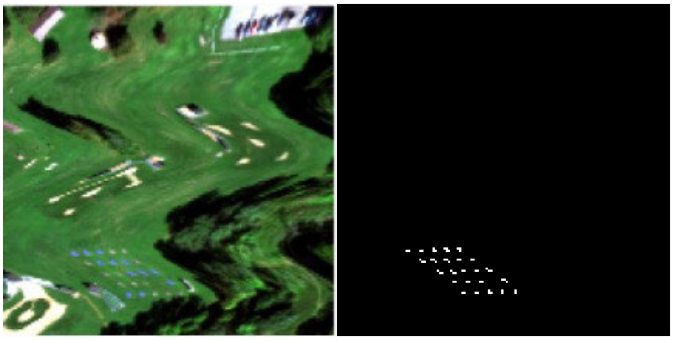
**Algorithm 1** GODSNCR for Target Detection

**Input**: HSI $\boldsymbol{X} \in \mathbb{R}^{L\times N}$, target spectra $\boldsymbol{S}$, parameter min_*cluster_size*, min_*samples*, number of background dictionary atoms *bg_num*, $\lambda$, sparsity level $K$.
**Output**: detection result $D_{\text{GODSNCR}}(x)$.
**Initialization**: dim=30, del_rate=0.05, $t_{\max} = 100$, $\rho = 0.05$, $\epsilon_0 = 10^{-2}$, $\boldsymbol{\alpha} = \boldsymbol{0}$, $\boldsymbol{Z} = \boldsymbol{0}$, $\boldsymbol{W} = \boldsymbol{0}$.
**Clustering**:
  Applying the PCA algorithm to reduce the dimension of $\boldsymbol{X}$ and obtain $\boldsymbol{X}_{\text{reduce}}$.
  Applying the HDBSCAN algorithm to cluster the $\boldsymbol{X}_{\text{reduce}}$ data and obtain clusters.
  Select the corresponding *bg_num* elements in the cluster as the unpurified overcomplete dictionary, and combine $\boldsymbol{S}$ to form a joint overcomplete dictionary $\boldsymbol{A}$.
  According to the del_rate, delete some atoms that are close to the prior target spectra $\boldsymbol{S}$ in the unpurified overcomplete dictionary to obtain the purified background dictionary $\boldsymbol{A}_b$.
**Iteration**:
  **While** $t < t_{\max}$, repeat
    Update weight matrix $\boldsymbol{\alpha}$ by Eq.(8).
    Calculate the tolerance $\epsilon$.
    **If** $\epsilon < \epsilon_0$
      **break**.
    **End if**.
    Update the variable $\boldsymbol{Z}$ by Eq.(9).
    Calculate the Lagrange multiplier $\boldsymbol{W}$ by Eq.(10).
  **End While**.
**Detection**:
  **For** each pixel $x$ in $\boldsymbol{X}$
    Applying the OMP algorithm to obtain the sparse representation coefficients of test pixel x using dictionary $\boldsymbol{A}$.
    Calculate $r_b$ and $r_t$ by Eq.(12) and Eq.(14).
    For each pixel $x$, calculate the residual by Eq.(15).
  **End for**.

*1) Mosaic Dataset:* This dataset [53] was collected from the SpecTIR Hyperspectral Aerial Experiment in 2012. The sensor ProSpecTIR captured an image scene of a park situated near a town, located south of Rochester, USA. The scene is depicted in Fig. 2(a). It consists of 360 spectral bands with a spectral resolution of 5 nm. The experimental area in this dataset consists of 150 × 150 pixels. The detection target consists of 60-pixel panels, with the ground truth presented in Fig. 2(b).

*2) RIT Campus Dataset:* It [54] was collected from the Rochester Institute of Technology campus by the sensor ProSpecTIR, as shown in Fig. 2(a). It contains 360 bands with a spectral resolution of 5 nm. The scene with 200 × 200 pixels is taken as the experimental area in this dataset. The detection target in this dataset consists of 80-pixel red panels, with the ground truth presented in Fig. 3(b).

*3) AVIRIS Dataset:* It [55] was collected from the San Diego airport by the sensor AVIRIS, as shown in Fig. 4(a). The AVIRIS dataset contains 189 bands, each with a spectral resolution of 10 nm. The Scene with 100 × 100 pixels is taken as the experimental area in this dataset. The detection target consists of 58-pixel planes, with the ground truth presented in Fig. 4(b).



(a) (b)

Fig. 2. Mosaic dataset (a) image scene and (b) ground truth.



(a) (b)

Fig. 3. RIT Campus dataset (a) image scene and (b) ground truth.



(a) (b)

Fig. 4. AVIRIS dataset (a) image scene and (b) ground truth.

## B. Experimental Settings

*1) Comparison Methods and Performance Evaluation Settings:* In the comparison experiments, we select several typical and advanced hyperspectral detection algorithms that are constructed based on representation learning and deep learning to compare with our proposed algorithm. The comparison algorithms are as follows: STD [25], SRBBH [27] detector, CSCR [26] detector, HTD framework using a deep

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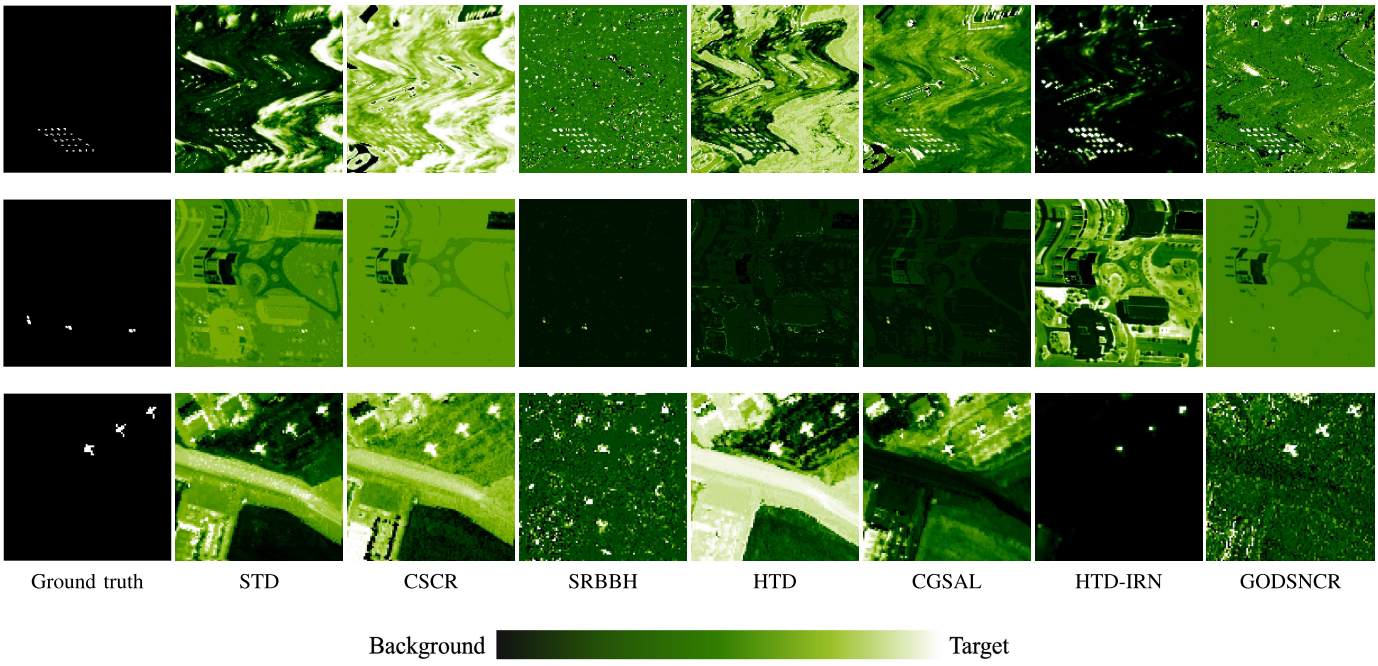



Fig. 5. Comparison of depth maps of target detection results in three groups of experimental data (from top to bottom are the results of Mosaic, RIT Campus, and AVIRIS datasets).

network with data augmentation [29], CGSAL model [28], and HTD-IRN [37].

Three reliable evaluation criteria, namely, receiver operating characteristic (ROC) curves [56], [57], area under ROC curves (AUCs) values [58], and the target-background separation graph [59], [60], are employed to assess the detection performances both qualitatively and quantitatively. This allows for accurate analysis and comparison of the detection efficacy of various target detectors.

*2) Parameter Settings:* The parameters of our proposed GODSNCR algorithm include clustering parameters min_cluster_size and min_samples, number of background dictionary atoms bg_num, coefficient of collaborative term $\lambda$, sparsity level $K$, and the maximum number of iterations $t_{\max}$. In the final experimental results, after selecting appropriate clustering parameters for the three groups of data (set min_cluster_size to 8 and min_samples to 8 under the Mosaic dataset, respectively; set min_cluster_size to 4 and min_samples to 8 under the other two datasets). We uniformly set bg_num to 450, $\lambda$ to 0.005, $K$ to 10, and $t_{\max}$ to 100 for all datasets. Specifically, we select two prior target spectra in the Mosaic dataset, and three prior target spectra in both the RIT campus dataset and the AVIRIS dataset. For the other comparison algorithms, parameter Settings are set according to the recommendations of the original papers. The inner and outer window sizes (OWR, IWR) are set uniformly for a single dataset: (7, 5) for the Mosaic dataset; (13, 21) for the RIT Campus dataset; and (7, 17) for the AVIRIS dataset. In particular, for the CGSAL algorithm, the background is known from the original text. Therefore, here we adopt the prior knowledge provided in the CGSAL paper as the background categories and conduct experiments by selecting one sample from each category in the purified background dictionary $\boldsymbol{A}_b$ constructed in this article. For the HTD-IRN algorithm, we conduct multiple experiments on the two parameters, $m$ and $\eta_1$, as mentioned in the original paper, across three datasets. We select values for $m = 30$ and $\eta_1 = 100$, which exhibit favorable performance in all three datasets, for a unified experiment.

### C. Detection Performance

The depth maps of target detection results for the datasets are shown in Fig. 5.

The depth map of the target detection results corresponds to the detection result. Specifically, for a pixel in the image, darker colors indicate a higher probability that the pixel corresponds to the background, and lighter colors indicate a higher probability that it corresponds to a target. Therefore, the depth map of the target detection results can intuitively show the background suppression effect and the target enhancement effect of the detection results.

From Fig. 5, it is noticeable that for the Mosaic dataset, the STD, CSCR, and HTD algorithms exhibit the ability to enhance a majority of the targets. But in comparison to the proposed GODSNCR algorithm, these algorithms exhibit comparatively weaker background suppression performance. Specifically, they struggle with suppressing the tree background in the image, leading to a significant number of false alarms. The CGSAL algorithm exhibits subpar performance in this dataset, due to ineffective target segmentation and noise reduction. It not only produces obvious false alarms in the upper right background region of the image but also misses some targets. The SRBBH algorithm performs moderately well in this dataset, with relatively uniform background

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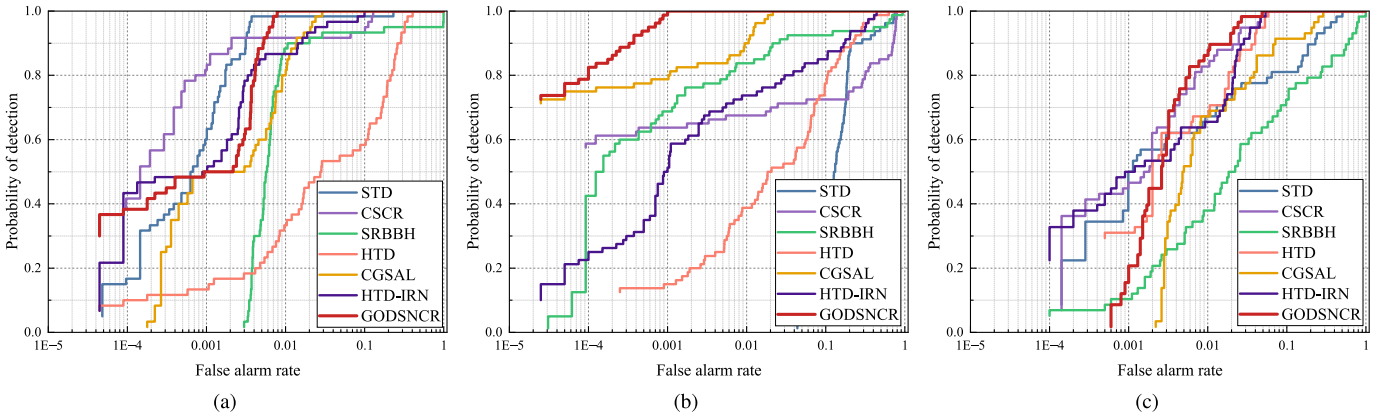



Fig. 6. ROC curves for three datasets. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

suppression throughout the whole image. However, false alarms are scattered throughout the image, which negatively affect the detection performance. The HTD-IRN algorithm performs relatively well in this dataset, especially in terms of background suppression, achieving decent results. However, it shows a relatively concentrated false alarm in some places in this dataset. The proposed GODSNCR algorithm exhibits better background suppression performance in this dataset due to the construction of a purer background dictionary in this dataset, particularly for the tree background in the lower right part of the image. This algorithm is capable of precisely locating the true target positions.

From Fig. 5, it can be observed that, except for the STD and the HTD-IRN algorithm, all other algorithms exhibit strong background suppression performance in the RIT Campus dataset. However, the CSCR algorithm tends to miss detecting the targets, and the SRBBH algorithm produces false alarms. Therefore, it is evident that the first four algorithms exhibit mediocre detection performance in this dataset. The CGSAL algorithm and the proposed GODSNCR algorithm achieve significant background suppression with excellent results, as they are less affected by noise interference in this dataset. In particular, the GODSNCR algorithm exhibits a higher detection rate for both target boundary pixels and overall targets, showing significant superiority in target detection.

Based on the results illustrated in Fig. 5, it is evident that the SRBBH algorithm performs poorly in the AVIRIS dataset compared to the other algorithms. As there is no procedure for constructing a pure background dictionary, the SRBBH algorithm, despite delivering relatively consistent background suppression, led to sporadic false alarms throughout the entire image and did not establish a clear differentiation between targets and the background. Although the STD, HTD, and CGSAL algorithms have only moderate background suppression effects, they exhibit significant highlights on the target, ensuring a certain level of detection capability. Specifically, the STD and HTD algorithms exhibit inadequate background suppression effects on the lower left house background, and the CGSAL algorithm shows insufficient suppression effects on the runway background around the aircraft. Compared to other algorithms, the HTD-IRN algorithm demonstrates the strongest background suppression capability in this dataset. However, it lacks the capability to completely detect the contour pixels around the target. The CSCR algorithm and the proposed GODSNCR algorithm both exhibit excellent background suppression abilities, and the enhancement of the target is more significant. They can locate the real target position well and have good detection performance. In comparison, the proposed GODSNCR algorithm particularly excels in background suppression, thereby reducing the number of false alarms in the lower left image zone and showing the best detection performance.

To conduct a more comprehensive evaluation of the GODSNCR algorithm, this article plots the ROC curves of each algorithm under three datasets, as shown in Fig. 6. The ROC curve of the GODSNCR algorithm is depicted as the red curve in the figure. Regardless of the dataset, the curve consistently occupies the upper-left part of the graph. Especially for the RIT Campus data, the GODSNCR algorithm demonstrates a superior detection rate compared to other algorithms at multiple false alarm rates. The SRBBH and HTD algorithms display inferior performance in the Mosaic dataset, evincing lower detection rates—an observation consistent with the conclusion deduced from the depth map visualization. Despite the proposed algorithm not being consistently superior to other algorithms when the false alarm rate is below 0.01, it demonstrates a detection rate highly approximating 1 when the false alarm rate surpasses 0.01, and outperforms other algorithms in terms of performance. Regarding the AVIRIS dataset, the performance of the SRBBH algorithm is notably weak in terms of detection, with its curve mostly trailing behind those of the other algorithms, which exhibits comparable performance. When the false alarm rate surpasses 0.01, the performance of the GODSNCR algorithm in detection becomes significantly better than that of other algorithms.

To more accurately evaluate the performance of the GODSNCR algorithm in the case of curve crossing, we calculate the area enclosed by the ROC curve, i.e., we obtain the AUC values as shown in Table I, and plot them as histograms as shown in Fig. 7. From the results in Table I and Fig. 7, the GODSNCR algorithm achieves the optimal detection results among all three datasets and is more stable compared to

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TABLE I
AUC VALUES OF DIFFERENT TARGET DETECTORS FOR THREE DATASETS

| Dataset | STD | CSCR | SRBBH | HTD | CGSAL | HTD-IRN | GODSNCR |
|---|---|---|---|---|---|---|---|
| Mosaic | 0.9929 | 0.9892 | 0.9341 | 0.8970 | 0.9944 | 0.9930 | **0.9977** |
| RIT Campus | 0.9375 | 0.8474 | 0.9467 | 0.9244 | 0.9978 | 0.9549 | **0.9999** |
| AVIRIS | 0.9440 | 0.9893 | 0.8467 | 0.9880 | 0.9668 | 0.9898 | **0.9942** |

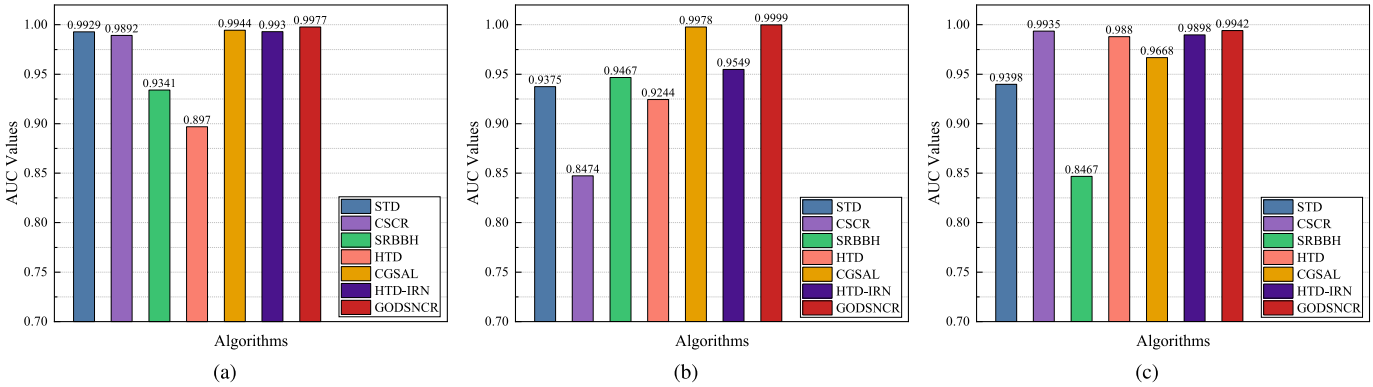



Fig. 7. AUC values for three datasets. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

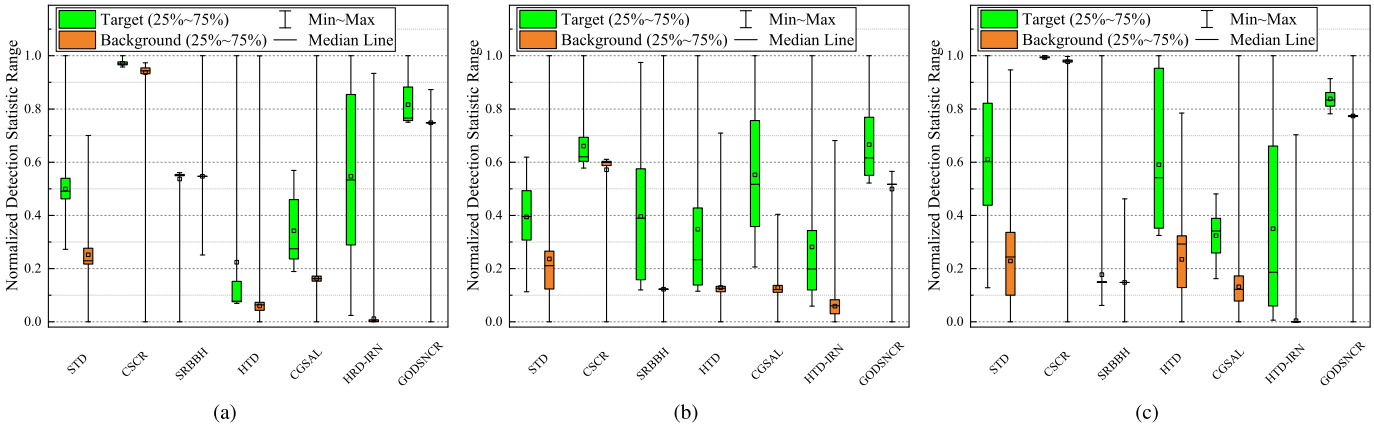



Fig. 8. Target-Background separability maps of different detectors for three datasets. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

other algorithms. It can be seen that the GODSNCR algorithm compensates the disadvantages of other local dual window approaches to some extent, and shows the superior performance of the overcomplete dictionary construction of the sparse representation and the consistency constraint of the collaborative representation on the detection results.

To further compare and analyze the ability of these algorithms for target-background separation during detection, the target-background separation is plotted for comparison in Fig. 8. The separability map illustrates the distribution characteristics of the target pixels (in green) and the background pixels (in orange), focusing on the main body (25%–75%), to assess their separability. A proficient detector is capable of effectively distinguishing the target from the backgrounds during the detection process. The comparison of the target-background separability maps illustrates that the STD and CSCR algorithms excel in detecting the target, but perform poorly in suppressing the background. Although both the SRBBH and the proposed algorithms significantly suppress background noise in three datasets, the SRBBH algorithm has a disappointing effect on enhancing the target detection, resulting in weak target-background separability. In contrast, the proposed algorithm effectively suppresses background noise while simultaneously enhancing target detection, resulting in significantly better target-background separability. The CGSAL algorithm's performance is inconsistent across the three datasets, favorably suppressing background noise in the Mosaic and RIT Campus datasets, but it only moderately suppresses background noise in the AVIRIS dataset. The HTD algorithm generally succeeds in suppressing background noise across multiple datasets. However, its ability to enhance the target is limited, leading to an overall mild separability, with particularly weak performance in the first two datasets. The HTD-IRN algorithm performs well in suppressing the background subject in three datasets, especially achieving strong background suppression effects in Mosaic and AVIRIS datasets. However, its ability to highlight the target is relatively weak, and the detection values of the

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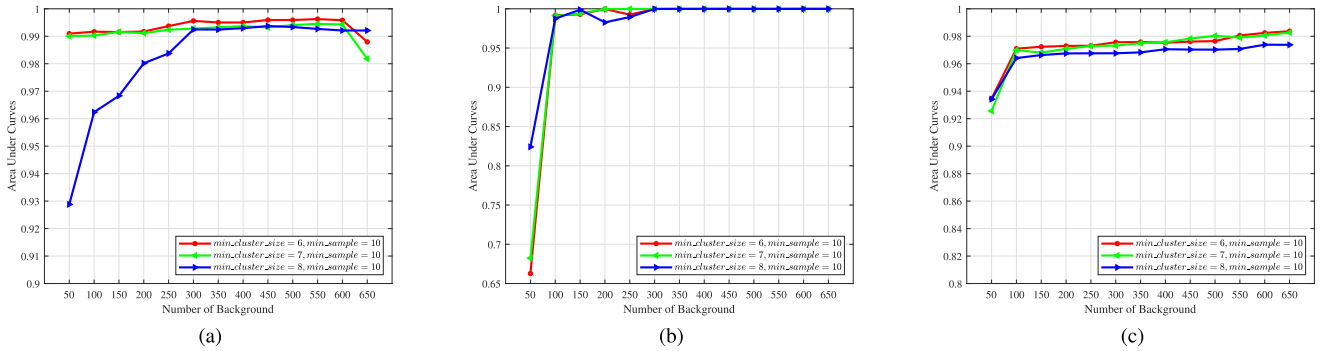



Fig. 9. Number of dictionary atoms under three datasets, results of analysis of parameter bg_num. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

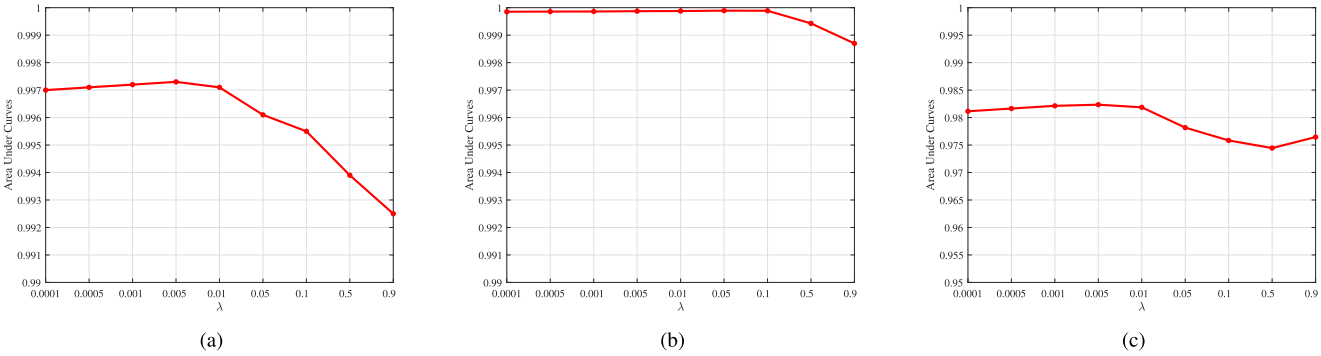



Fig. 10. Results of the parametric analysis of the coefficient of the collaborative term under three datasets. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

target are within a larger range. A comprehensive comparison of methods illustrates that the proposed method not only successfully suppresses background noise within a very narrow range across multiple datasets but also excels in enhancing the target, showcasing its unique advantages in target-background separation ability, and emphasizing its robustness.

### D. Parametric Analysis

The sensitivity analysis of the parameters of the GODSNCR algorithm is performed here in the order of the number of dictionary atoms, coefficient of collaborative term, sparsity level, and clustering parameters, and the evaluation index is the AUC value.

*1) Analysis of the Number of Background Dictionary Atoms:* Initially, we analyze the number of atoms in the background dictionary, setting $\lambda = 0.001$, and Sparsity $=$ 10 based on experience. For the experiment, three more general grouping clustering parameters are selected. From the results of the parameter analysis in Fig. 9, it is evident that the detection accuracy of the algorithms all show a tendency first to increase and then stabilize or slightly decrease as the number of dictionary atoms increases. This also indicates that the process of the target sparse representation dictionary from nonovercomplete to overcomplete provides more selectable basic vectors to participate in linear combinations, which increases the accuracy of sparse representation. However, as the number of dictionaries continues to increase, the possibility of introducing noise into the collaborative background representation inevitably increases, which may result in a decrease in detection accuracy. For the sake of robustness, the parameter that performs well under all datasets is chosen here: the number of background dictionary atoms is 450 for the subsequent parameter analysis experiments.

*2) Analysis of the Coefficient $\lambda$:* For the coefficient $\lambda$, we choose to set $\lambda$ as 0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, and 0.9 to test the detection accuracy of the GODSNCR algorithm by the coefficients $\lambda$ for different $\lambda$ under three sets of experimental data. The results are shown in Fig. 10. It can be seen from Fig. 10 that the detection accuracy of the proposed algorithm initially increases before decreasing as the collaborative term coefficient parameter is increased. This trend may occur due to an excessively large collaborative term coefficient parameter, causing an excessive collaborative constraint that destabilizes the model's solution. Specifically, for Fig. 10(b), a larger collaborative term parameter emphasizes collaboration more strongly, tending to utilize more dictionary atoms for collaborative representation. On the other hand, a smaller collaborative term parameter weakens this penalty, potentially leading to more independent changes. In cases where there are a greater number of background atoms and the background dictionary is pure, both using more atom representations and allowing more independent changes can achieve better detection performance. Although the detection accuracy of the model fluctuates somewhat with the change of the coefficient parameter of the collaborative term, the overall detection accuracy of the model is maintained at a relatively high level, indicating the robustness of the proposed algorithm. Here, we select the value with good performance under multiple datasets and set $\lambda$ as 0.005 for the subsequent parameter analysis experiment.

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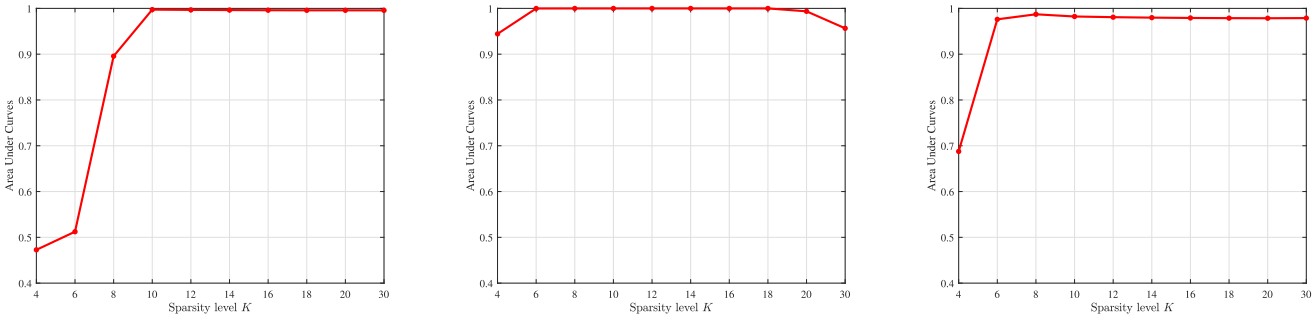



Fig. 11. Results of sparsity level parameter analysis under three datasets. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

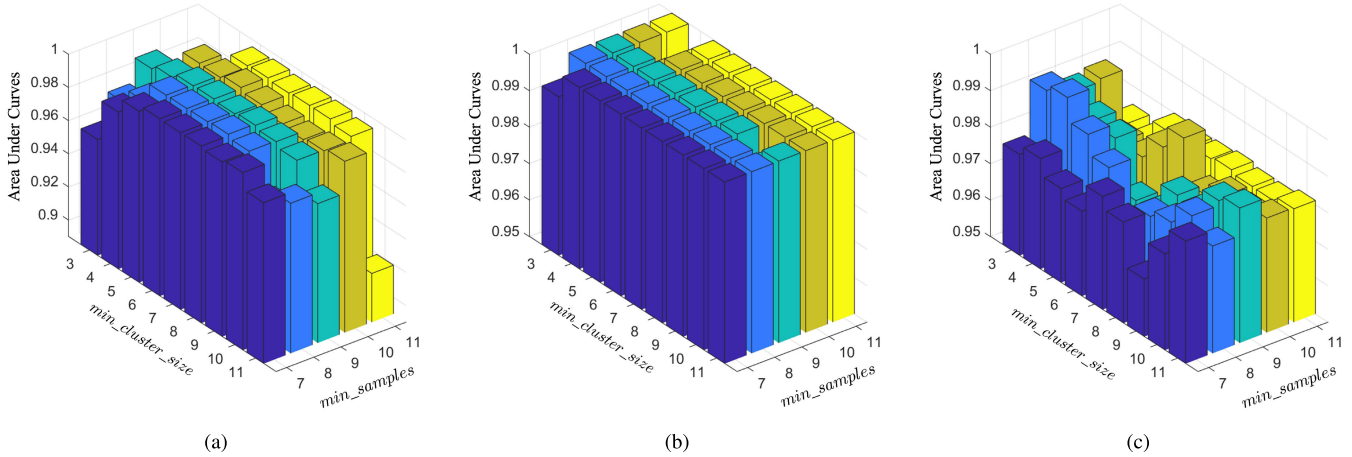



Fig. 12. Cluster parameter analysis results under three datasets. (a) Mosaic. (b) RIT Campus. (c) AVIRIS.

*3) Analysis of Sparsity Level $K$:* We set sparsity level $K$ to 4, 6, 8, 10, 12, 14, 16, 18, 20, and 30 for experimental analysis. It can be seen from Fig. 11 that with the increase of sparse level, the AUC value shows a tendency first to increase and then stabilize or slightly decrease with the increase of the number of sparse levels. The main reason here is that at the beginning, as the sparsity level increases, the number of nonzero elements of the sparse vector increases, and the target spectral information in the dictionary can be better used to represent the target, thus increasing the detection accuracy. Later, with the increase of sparsity level to a certain extent, other information will be involved in the representation process to represent the target, resulting in a decrease in accuracy. Specifically, for Fig. 11(b), a smaller sparsity already suffices to represent targets in this dataset. Therefore, after reaching a sparsity of 6, detection performance reaches a satisfactory level. Subsequent trends follow those observed in other datasets, stabilizing or slightly decreasing detection accuracy due to excessive sparsity introducing noise. From the experimental results, it is found that when sparsity level $K$ is 10, the results under multiple datasets are relatively robust. Here, sparsity level $K$ is selected as 10 for the following experiments.

*4) Analysis of Clustering Parameters:* For clustering parameters, min _cluster_size is selected as 3, 4, 5, 6, 7, 8, 9, 10, and 11, and min _samples is selected as 7, 8, 9, 10, and 11 for experimental analysis. It can be seen from Fig. 12 that min _cluster_size and min_sample have favorable effects when ranging between 4 and 9 and between 7 and 11, respectively. This observation aligns with the previous analysis of clustering parameter settings, which requires the selection of parameters that can effectively extract clusters with slightly fewer distribution numbers. Analysis of clustering parameters for the three datasets shows that the clustering algorithm utilized in this article is highly robust. When applied within a certain range of parameters, appropriate and reliable clustering outcomes can be realized. Consequently, it simplifies the process of constructing the dictionary and enhances the robustness of the overall algorithm.

### *E. Analysis of Time Cost*

In this section, in order to evaluate the efficiency of the proposed GODSNCR algorithm, we have tabulated the execution time (measured in seconds) of various comparison algorithms on three datasets, as shown in Table II. In the comparison, for the deep learning-based algorithms, we present both the training and testing times of the algorithms. Specifically for the GODSNCR algorithm, the time expended in the data preprocessing phase, which encompasses dictionary construction and preliminary steps, is outlined under the "Train/Data" category. Meanwhile, the time taken for subsequent detection processes is reported in the "Test" category.

For each detector, we conducted the experiment ten times under identical operating conditions. We then calculate the

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TABLE II
EXECUTION TIME (IN SECONDS) COMPARISON OF DIFFERENT DETECTORS FOR THREE DATASETS

| Dataset | Process | STD | CSCR | SRBBH | CGSAL | HTD | HTD-IRN | GODSNCR |
|---|---|---|---|---|---|---|---|---|
| Mosaic | Train/Data | $8.48 \pm 0.25$ | $5.88 \pm 0.06$ | $14.21 \pm 0.08$ | $10.05 \pm 0.26$ | $356.06 \pm 6.85$ | $323.43 \pm 2.54$ | $2.69 \pm 0.13$ |
| | Test | | | | | $8.64 \pm 1.10$ | $5.84 \pm 0.64$ | $20.79 \pm 0.19$ |
| RIT Campus | Train/Data | $61.85 \pm 0.27$ | $179.44 \pm 0.74$ | $153.87 \pm 1.17$ | $21.79 \pm 1.79$ | $415.7 \pm 6.18$ | $465.51 \pm 1.77$ | $6.42 \pm 0.49$ |
| | Test | | | | | $13.28 \pm 1.26$ | $9.56 \pm 0.69$ | $40.26 \pm 0.04$ |
| AVIRIS | Train/Data | $9.52 \pm 0.22$ | $24.62 \pm 0.99$ | $13.36 \pm 0.33$ | $2.85 \pm 0.12$ | $342.78 \pm 7.81$ | $197.08 \pm 4.56$ | $1.35 \pm 0.05$ |
| | Test | | | | | $3.67 \pm 0.13$ | $5.18 \pm 0.42$ | $7.54 \pm 0.06$ |

mean and standard deviation of the running times to evaluate the execution time performance of the various detectors. From the running time, it can be seen that the efficiency of the local dual window approach, such as STD, CSCR, and SRBBH, is influenced not only by the data scale but also significantly by the window size. Although the Mosaic dataset has a larger data scale than the AVIRIS dataset, the local window used is smaller, resulting in a smaller number of atoms in the constructed local background dictionary, and ultimately a smaller time cost than the AVIRIS dataset. The RIT Campus dataset, being the largest and using larger windows, has the highest time cost. The CGSAL algorithm directly uses the dictionary samples corresponding to the prior background for experimentation, hence the time cost is relatively small across all three datasets. The deep learning-based HTD and HTD-IRN algorithms have a larger time expenditure during the training phase but are less time-consuming in the testing phase. The GODSNCR algorithm has a relatively low data preprocessing cost in all three datasets, within an acceptable range. In terms of time expenditure during the detection process, it has a certain advantage over methods based on representation learning with larger local windows. For example, its time cost on the RIT Campus and AVIRIS datasets is less than that of the STD, CSCR, and SRBBH algorithms, which use local windows. Overall, the GODSNCR algorithm demonstrates relatively stable detection efficiency across various datasets, ensuring a certain level of detection efficiency.

## IV. CONCLUSION

Although representation learning-based methods have made breakthroughs in HTD, there remain numerous difficulties and challenges. This article proposes the GODSNCR for target detection. To fulfill the requirement of dictionary overcompleteness in sparse representation, we utilize a hierarchical density clustering algorithm to extract dictionary atoms and construct a joint overcomplete dictionary. To address the problem of the collaborative representation coefficient being negative and failing to meet physical meaning, we add a nonnegative constraint and a "sum to one" constraint to the coefficient matrix. To address the challenges of target noise or anomalous influence in the outer window due to the local dual window approach, we use the global background dictionary obtained by hierarchical density clustering instead of the background dictionary constructed by the local dual window approach. Extensive experiments are conducted using three real HSIs to validate the effectiveness of the GODSNCR method. The performance of the GODSNCR method is compared with other detectors based on representation learning and deep learning. Experimental results show that GODSNCR achieves excellent performance in HTD, outperforming other algorithms based on representation learning and deep learning. Future research will be dedicated to finding a more appropriate dictionary extraction technique that can produce a purer background dictionary, which can enhance the performance of the algorithm under low false alarm rates. In addition, we will concentrate on improving the performance of subpixel target detection by addressing problems such as difficulty in suppressing background components and estimating target components due to the highly mixed target background in subpixel target detection tasks.

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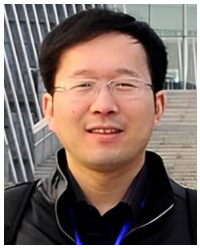
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